

CHAPTER 2

Review of Related Literature and Studies

This study presents the review of related literature and studies that serve as the theoretical and conceptual foundation of the proposed Maisagip, a mobile-responsive web application that utilizes computer vision and a hybrid deep learning framework for the post-harvest quality inspection of harvested white corn and yellow sweet corn ears. It discusses existing literature on computer vision in agriculture, deep learning, hybrid deep learning frameworks, EfficientNetV2, YOLOv11s, Rule-Based Decision Support Systems, and the Philippine National Standards (PNS/BAFS) for corn quality assessment. This chapter also examines both foreign and local studies related to agricultural image analysis, quality inspection, object detection, and artificial intelligence-based decision support systems. Furthermore, the reviewed literature and studies are analyzed to identify existing research gaps and establish the relevance of the proposed study. These serve as the foundation for the development of Maisagip and support the proposed approach in providing an objective, efficient, standardized, and accessible system for post-harvest corn quality assessment through a mobile-responsive web application.

Research Paradigm

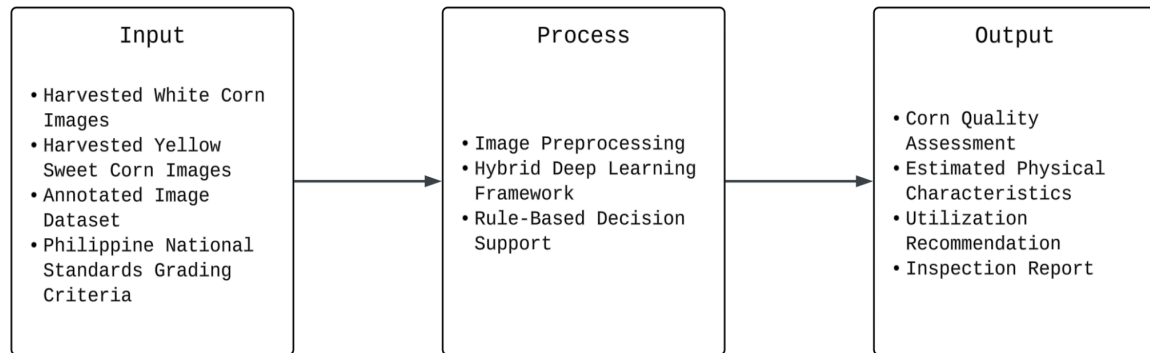


Figure 1 Research Paradigm of the Study

The research paradigm illustrates the Input–Process–Output (IPO) framework of the proposed Maisagip system. The input consists of harvested white corn ear images, harvested yellow sweet corn ear images, an annotated image dataset, and the Philippine National Standards (PNS/BAFS) grading criteria used to support quality assessment. During the process, the collected images undergo image preprocessing before being analyzed using a hybrid deep learning framework. The framework performs corn variety classification, defect detection, ear size estimation, and kernel completeness estimation. The resulting information is then processed through a rule-based decision support module that determines the appropriate quality grade, generates utilization recommendations, and prepares the inspection results.

The output of the system consists of an automated corn quality assessment, estimated physical characteristics like ear size and kernel completeness, utilization recommendations, and an automated inspection report. The system is implemented as a mobile-responsive web application, allowing users to access the inspection system through different platforms. Through this approach, the proposed system aims to provide an objective, consistent, and efficient method for post-harvest quality inspection of harvested white corn and yellow sweet corn while reducing dependence on manual grading.

Review of Related Literature

Computer Vision in Agriculture

Computer vision has become one of the most significant technologies in modern agriculture because it enables automated inspection, monitoring, and quality assessment of agricultural products through digital image analysis. By allowing computers to interpret and analyze visual information, computer vision reduces the need for manual inspection while improving the speed, accuracy, and consistency of agricultural operations. Unlike traditional inspection methods that rely heavily on human judgment, computer vision provides objective and non-destructive evaluation of agricultural commodities based on their observable physical characteristics.

According to Wang (2026), conventional grain quality inspection relies heavily on manual visual assessment, making the process labor-intensive, subjective, and susceptible to inconsistencies. The review explains that advances in machine vision, particularly deep learning techniques, have significantly improved automated grain quality inspection by enabling computer vision systems to extract complex visual features without relying on handcrafted image-processing methods. These technologies allow accurate evaluation of grain color, texture, morphology, and visible surface defects while providing efficient, non-destructive inspection suitable for post-harvest quality assessment of agricultural products intended for human consumption.

Xue et al. (2026) reviewed the applications of image recognition technologies in intelligent agricultural engineering. The authors explained that computer vision has been widely

applied in agricultural tasks such as crop identification, quality assessment, disease detection, yield estimation, and post-harvest inspection. The review highlighted that deep learning-based image recognition systems improve the efficiency, consistency, and accuracy of agricultural inspection while reducing reliance on manual evaluation. The authors also emphasized the importance of developing intelligent systems that integrate multiple artificial intelligence techniques to support more comprehensive agricultural applications.

Cao et al. (2025) reviewed the applications of computer vision and deep learning in modern agriculture. The authors explained that image-based artificial intelligence has been widely adopted for agricultural tasks such as crop monitoring, disease detection, plant growth assessment, and quality evaluation. Their review emphasized that integrating computer vision with deep learning enables automated and accurate agricultural inspection, reduces dependence on manual evaluation, and supports more efficient decision-making in agricultural production. These findings highlight the growing potential of intelligent computer vision systems for improving quality inspection and post-harvest management.

The findings of these studies provide a strong theoretical foundation for the present research. Similar to previous computer vision applications, the proposed Maisagip system utilizes digital image analysis to automate the post-harvest quality inspection of harvested white corn and yellow sweet corn ears. Rather than relying on subjective manual grading, the system analyzes observable external characteristics through computer vision to provide objective, efficient, and standardized quality assessment. Furthermore, the proposed study extends previous applications by integrating multiple inspection functions, including corn variety classification, visible defect detection, quality grading, physical characteristic estimation, recommendation

generation, and automated inspection reporting within a single web-based intelligent system using a hybrid deep learning framework.

Deep Learning in Agriculture

Deep learning has become one of the most influential technologies in artificial intelligence because of its ability to automatically learn complex patterns and features from large amounts of data. Unlike traditional machine learning techniques that require manual feature extraction, deep learning models can automatically identify meaningful visual features directly from images, making them highly effective for computer vision applications. This capability has significantly improved the performance of automated image recognition systems across various fields, including healthcare, manufacturing, transportation, and agriculture.

In the agricultural sector, deep learning has transformed the way crops and agricultural products are analyzed by enabling intelligent systems to perform tasks such as crop classification, disease detection, quality grading, yield estimation, and post-harvest inspection with minimal human intervention. According to Liao et al. (2026), deep learning-based image recognition technologies have significantly advanced food quality assessment by enabling automated image analysis with minimal manual feature extraction. The authors explained that deep learning models, including convolutional neural networks (CNNs) and transformer-based architectures, can effectively learn complex visual features from food images to support tasks such as quality evaluation, defect detection, classification, and grading. The review emphasized that these technologies improve the accuracy, efficiency, and reliability of food inspection while providing greater adaptability to diverse food products and inspection environments.

A review by Nawaz et al. (2025) presented a comprehensive survey of more than 200 research works on artificial intelligence techniques for agriculture, covering conventional machine learning, deep learning, vision transformers, and vision-language foundation models. The review discussed the application of these techniques in agricultural tasks such as crop disease detection, object detection, crop monitoring, livestock health assessment, and aquatic species monitoring. The authors highlighted that deep learning and computer vision have significantly improved the accuracy and efficiency of agricultural analysis while also identifying challenges related to data quality, model generalization, and deployment in real-world farming environments. They further emphasized the need for multimodal data integration, efficient edge-device deployment, and adaptable AI models to support future intelligent agricultural systems.

Furthermore, Kamilaris and Prenafeta-Boldú (2023) reviewed the application of deep learning techniques in agriculture and highlighted their growing importance in agricultural image analysis. The authors explained that deep learning models have been widely applied to tasks such as image classification, object detection, semantic segmentation, and quality assessment, enabling more accurate and automated analysis of agricultural images. The review emphasized that deep learning can effectively learn complex visual features from image data, supporting improved performance across a wide range of agricultural applications. These findings demonstrate the potential of deep learning to enhance the efficiency, accuracy, and reliability of intelligent agricultural inspection systems.

The rapid advancement of deep learning has also encouraged the development of more specialized architectures designed to improve both performance and computational efficiency. Recent studies have introduced lightweight and optimized models capable of performing

complex image recognition tasks while reducing processing time and hardware requirements. These improvements have made deep learning more practical for real-world agricultural applications, particularly in post-harvest quality inspection, where fast and accurate analysis is essential.

The findings of these studies provide a strong theoretical foundation for the present research. The proposed system utilizes deep learning to automate the post-harvest quality inspection of harvested white corn and yellow sweet corn ears using RGB images. By replacing subjective manual inspection with intelligent image analysis, the system aims to provide more objective, accurate, and consistent quality assessment. Furthermore, the study adopts a Hybrid Deep Learning Framework that combines specialized models to perform multiple inspection tasks within a unified system, allowing more comprehensive and efficient quality evaluation than conventional single-model approaches.

Hybrid Deep Learning Framework

Recent advancements in artificial intelligence have led to the development of hybrid deep learning frameworks, which integrate two or more deep learning models and intelligent techniques to solve complex problems more effectively than a single model. Instead of relying on one algorithm to perform all tasks, a hybrid framework combines the strengths of different models, allowing each component to perform a specialized function while working together within a unified system. This approach improves overall accuracy, efficiency, robustness, and decision-making, particularly in applications that require multiple image analysis tasks.

Cao et al. (2025) demonstrated that hybrid deep learning approaches have gained significant attention in agricultural image analysis because they improve feature extraction and enable intelligent systems to perform multiple related tasks, such as image classification, object detection, quality assessment, and decision support. The authors explained that integrating complementary deep learning models allows agricultural systems to achieve higher accuracy and better generalization than conventional single-model approaches. Hybrid frameworks also reduce computational redundancy by allowing each model to focus on its specific task, resulting in more efficient processing and improved inspection performance.

Song et al. (2023) emphasized that combining different deep learning architectures has become an effective strategy for addressing complex agricultural image analysis problems. The researchers explained that no single model is capable of performing all inspection tasks with optimal performance. Image classification models excel at recognizing categories, while object detection models are more effective in identifying and localizing visible defects. Integrating these specialized models within a hybrid framework enables intelligent agricultural systems to generate more comprehensive inspection results while maintaining high accuracy and computational efficiency.

Liao et al. (2026) further explained that modern food quality inspection systems increasingly combine multiple artificial intelligence techniques to improve reliability and support automated decision-making. Rather than depending solely on deep learning predictions, integrating additional decision-support mechanisms allows inspection systems to produce more meaningful outputs, including quality grading, product recommendations, and automated reporting. Such integrated approaches make intelligent inspection systems more practical for

real-world food quality assessment because they provide users with interpretable and actionable results instead of simple classification outputs.

The application of hybrid deep learning frameworks has become increasingly important in agricultural quality inspection, where multiple tasks must often be performed simultaneously. For harvested corn, quality assessment involves not only identifying the crop variety but also detecting visible defects, evaluating physical characteristics, assigning quality grades, and generating recommendations based on established grading standards. Performing these tasks using separate independent systems increases computational complexity, processing time, and implementation cost. A hybrid framework addresses these limitations by integrating specialized models into a single intelligent system capable of producing comprehensive inspection results efficiently.

EfficientNetV2 for Image Classification

Image classification is one of the fundamental tasks in computer vision, wherein a deep learning model automatically identifies the category or class of an object based on its visual features. In agricultural applications, image classification is widely used for crop identification, fruit recognition, seed classification, disease diagnosis, and quality assessment. The rapid advancement of convolutional neural networks (CNNs) has significantly improved the accuracy of image classification by enabling models to automatically learn complex image features such as color, texture, shape, and structural patterns without manual feature extraction.

In accordance with Tan and Le (2021), EfficientNetV2 is an improved version of the EfficientNet architecture designed to achieve higher classification accuracy while significantly reducing training time and computational cost. The researchers introduced fused mobile inverted bottleneck convolution (Fused-MBConv) layers together with progressive learning strategies to improve feature extraction and accelerate model training. Experimental results demonstrated that EfficientNetV2 outperformed previous EfficientNet models and several state-of-the-art convolutional neural networks by achieving better accuracy with fewer computational resources. These improvements make EfficientNetV2 suitable for applications that require both high performance and computational efficiency.

The findings support the adoption of EfficientNetV2 as the image classification component of the proposed Hybrid Deep Learning Framework. In the present study, EfficientNetV2 serves as the initial stage of the inspection process by identifying the captured image as either white corn or yellow sweet corn. Since these two corn varieties may exhibit variations in color intensity, kernel arrangement, ear size, maturity, lighting conditions, viewing angles, and visible external characteristics, a robust image classification model is necessary to ensure reliable identification. The classification result is then combined with the outputs of the defect detection model and the rule-based decision support module to generate a comprehensive post-harvest quality assessment. Through its high accuracy, lightweight architecture, and efficient feature extraction capability, EfficientNetV2 contributes to the overall reliability and effectiveness of the proposed Maisagip system.

YOLOv11s for Object Detection

Object detection is a computer vision task that enables a system to identify, locate, and classify multiple objects within an image simultaneously. Unlike image classification, which predicts only the overall category of an image, object detection determines the exact location of specific objects by drawing bounding boxes around them while assigning the appropriate class labels. In agriculture, object detection has become an essential technology for identifying crop diseases, detecting pests, monitoring plant growth, counting fruits, and inspecting the quality of agricultural products through automated image analysis.

Recent studies have demonstrated the effectiveness of YOLO-based models in agricultural quality inspection. Xia et al. (2024) discussed the effectiveness of the YOLOv5 object detection model for identifying visible surface defects in maize seeds. Their findings showed that deep learning can accurately detect external defects while enabling fast and automated inspection, reducing the limitations of manual evaluation.

Yang et al. (2024) proposed an improved lightweight YOLOv7 model for maize kernel quality detection. The researchers enhanced the original YOLOv7 architecture to improve detection performance while reducing computational complexity, making the model more suitable for real-time agricultural applications. Their results demonstrated that the improved model achieved high detection accuracy and efficient inference speed, indicating its potential for automated maize quality inspection systems.

Furthermore, Xu et al. (2026) highlighted an enhanced object detection model, FCD-GELAN, for fresh corn appearance inspection by combining visible and near-infrared (Vis–NIR) image fusion. The proposed model achieved an mAP@0.5 of 90.2%, demonstrating

strong detection performance for identifying appearance defects in fresh corn. The study showed that integrating Vis–NIR image fusion with an improved deep learning architecture can enhance the accuracy and robustness of corn appearance inspection. These findings highlight the potential of advanced object detection techniques for supporting automated post-harvest quality assessment of agricultural products.

Although the reviewed studies utilized earlier YOLO architectures, the rapid evolution of the YOLO family has resulted in newer versions with improved detection performance. Among these, YOLOv11s is designed as a lightweight variant that provides an effective balance between detection accuracy, inference speed, and computational efficiency. Its architecture enables reliable detection of small and subtle defects while remaining suitable for deployment in web-based intelligent systems that require fast response times and efficient resource utilization.

The findings provide a strong foundation for the present study. In Maisagip, YOLOv11s serves as the object detection component of the proposed Hybrid Deep Learning Framework. The model is responsible for detecting visible external defects on harvested white corn and yellow sweet corn ears, including mold growth, insect damage, discoloration, deformities, missing kernels, and other observable physical defects. The detected defects are then forwarded to the Rule-Based Decision Support System, where they are evaluated together with the classification results and estimated physical characteristics to determine the corresponding quality grade and utilization recommendation. By integrating a lightweight yet highly accurate object detection model, the proposed system aims to provide efficient, reliable, and automated post-harvest quality inspection suitable for practical agricultural applications.

Rule-Based Decision Support System

A Rule-Based Decision Support System (RDSS) is an intelligent decision-making approach that applies predefined rules and domain knowledge to generate consistent and explainable recommendations. Unlike deep learning models that automatically learn patterns from data, rule-based systems use logical IF-THEN rules to transform model predictions into standardized decisions based on established criteria. Because the decision-making process follows explicit rules, the generated outputs are transparent, interpretable, and easier for users to understand.

Recent studies have highlighted the increasing importance of integrating rule-based reasoning with artificial intelligence to improve the transparency and reliability of intelligent decision support systems. The review by Kostopoulos et al. (2024), concluded that explainable AI-based decision support systems combine machine learning predictions with rule-based reasoning to provide decisions that are not only accurate but also understandable to users. The authors explained that rule-based methods, particularly production rule systems using IF-THEN rules, improve explainability by allowing users to trace how decisions are generated while maintaining strong predictive performance. They further emphasized that rule-based reasoning is especially valuable in applications that require decisions to follow established standards and expert knowledge.

A review by Pai et al. (2025) explained that rule-based models remain an important component of intelligent agricultural systems because they enhance the transparency, trustworthiness, and practical adoption of artificial intelligence. Their review highlighted that combining deep learning with rule-based reasoning allows agricultural systems to produce

decisions that are easier for farmers, researchers, and policymakers to interpret while maintaining high analytical performance. The authors also noted that explainable and rule-based approaches are becoming increasingly important in agricultural applications that involve quality assessment and decision support.

The findings provide a strong foundation for the present study, where the outputs generated by EfficientNetV2 and YOLOv11s are processed through a Rule-Based Decision Support System that applies the Philippine National Standards (PNS/BAFS) and other recognized agricultural grading references. Based on the detected corn variety, visible defects, kernel completeness, and ear size, the system automatically assigns the appropriate quality grade, generates utilization recommendations, and produces an inspection report. By integrating deep learning with rule-based reasoning, the proposed Hybrid Deep Learning Framework provides a more transparent, standardized, and practical approach to automated post-harvest corn quality inspection.

Philippine National Standards (PNS/BAFS) as the Basis for Corn Quality Assessment

Standardized grading provides an important role in post-harvest quality management because it promotes consistency, fairness, and objectivity in evaluating agricultural products. In the Philippines, the Bureau of Agriculture and Fisheries Standards (BAFS) develops the Philippine National Standards (PNS) to establish uniform quality requirements and grading systems for various agricultural commodities. These standards serve as official references for producers, traders, processors, and regulatory agencies in assessing product quality and ensuring that agricultural commodities meet market and consumer requirements.

For fresh sweet corn, PNS/BAFS 98:2022 - Sweet Corn - Product Standard - Grading and Classification provides standardized requirements for evaluating post-harvest quality. The standard states that harvested sweet corn should be properly developed and mature, capable of withstanding handling and transportation while maintaining satisfactory quality upon reaching its destination. It also identifies maturity indicators such as ear fill, silk drying, kernel development, kernel sweetness, and kernel tenderness. Furthermore, the standard classifies sweet corn into Extra Class, Class I, and Class II, based on observable quality characteristics, including ear uniformity, kernel development, trimming quality, and the allowable percentage of visible defects present on the cob.

The grading principles defined in PNS/BAFS emphasize the importance of evaluating externally observable characteristics, making them compatible with computer vision technologies. Observable attributes such as kernel completeness, ear size, discoloration, mold symptoms, insect damage, deformities, and other visible physical defects can be assessed through digital image analysis without damaging the harvested corn. These characteristics provide objective indicators that support consistent quality classification while reducing the subjectivity associated with manual inspection.

In addition to standards for fresh corn, the Philippines also implements PNS/BAFS 15:2018 - Corn (Maize) Grits - Grading and Classification, which establishes quality requirements for processed corn products. Although this standard applies specifically to corn grits rather than fresh corn ears, it demonstrates the country's commitment to maintaining standardized quality assessment throughout different stages of the corn value chain.

The findings provide an important foundation, where the Rule-Based Decision Support System adopts the grading principles and observable quality characteristics described in PNS/BAFS 98:2022, together with other recognized agricultural grading references, to support automated quality assessment. The outputs generated by EfficientNetV2 and YOLOv11s are evaluated using predefined decision rules based on observable characteristics, including visible defects, kernel completeness, ear size, and overall appearance. The system then automatically assigns the appropriate quality grade, generates utilization recommendations, and produces a comprehensive inspection report. By incorporating nationally recognized grading principles into the proposed Hybrid Deep Learning Framework, the study promotes a more objective, standardized, and explainable approach to automated post-harvest quality inspection of harvested white corn and yellow sweet corn.

Review of Related Studies

Foreign Studies

Yang et al. (2024) developed a non-destructive maize kernel defect detection system using hyperspectral imaging combined with a convolutional neural network (CNN) with attention mechanisms. The researchers evaluated different types of maize kernel conditions, including healthy, moldy, insect-damaged, broken, sprouted, and heat-damaged kernels. By integrating spectral and spatial attention modules into the CNN architecture, the proposed model achieved a classification accuracy of 94.56%, demonstrating improved performance compared with traditional machine learning approaches. The study concluded that deep learning combined with advanced imaging technologies can provide an accurate and non-destructive approach for

maize quality evaluation. This study is relevant to the present research because it demonstrates the capability of deep learning models in identifying visible quality problems in corn products. However, the study focuses only on individual maize kernels and requires hyperspectral imaging equipment, unlike Maisagip which aims to inspect whole corn ears using conventional RGB images suitable for practical agricultural applications.

Yang et al. (2024) proposed an improved lightweight YOLOv7 model for maize kernel quality detection. The researchers modified the original YOLOv7 architecture by introducing lightweight improvements to reduce computational complexity while maintaining detection accuracy. The developed model demonstrated improved detection performance and faster inference speed, making it more suitable for real-time agricultural inspection applications. The study highlighted the effectiveness of object detection algorithms in automatically locating and identifying maize quality issues. This research supports Maisagip by demonstrating the potential of YOLO-based models for agricultural defect detection. However, unlike the present study, the proposed system focuses only on maize kernels and performs a single detection task rather than integrating variety classification, defect detection, quality grading, and recommendation generation within one framework.

Xia et al. (2024) developed a YOLOv5-based computer vision system for detecting surface defects in maize seeds. The researchers created an image acquisition system and trained deep learning models to identify external defects affecting seed quality. The results showed that the improved YOLOv5 model achieved accurate defect detection while maintaining efficient processing performance. The researchers emphasized that deep learning-based inspection can reduce dependence on manual evaluation and improve the speed and consistency of agricultural quality assessment. This study is closely related to Maisagip because both utilize computer

vision and object detection for corn quality inspection. However, the study focuses on maize seeds rather than harvested corn ears and does not include automated quality grading or physical characteristic estimation.

Chen et al. (2026) introduced Corn-Net, a semantic segmentation-based system for post-harvest fresh corn sorting. The proposed model incorporated attention mechanisms and multi-scale feature extraction techniques to improve segmentation performance and identify important visual characteristics of fresh corn. Experimental results showed that Corn-Net achieved high segmentation accuracy while maintaining efficient inference speed for agricultural applications. The researchers concluded that semantic segmentation can support automated fresh corn sorting by providing accurate identification of corn regions and quality-related features. This study is relevant because it directly addresses post-harvest fresh corn inspection using artificial intelligence. However, the system primarily focuses on segmentation and sorting, while Maisagip extends the inspection process by combining classification, defect detection, quality grading, physical estimation, and recommendation generation.

Xu et al. (2026) developed the FCD-GELAN model for fresh corn appearance detection using visible and near-infrared (Vis-NIR) image fusion. The researchers enhanced the GELAN architecture with attention mechanisms to improve the detection of subtle surface defects in fresh corn. The proposed model achieved an mAP@0.5 of 90.2%, demonstrating strong detection capability and outperforming several existing object detection models. The study emphasized that combining advanced imaging techniques with deep learning can improve agricultural quality inspection accuracy. This research supports the present study by demonstrating the effectiveness of object detection models for fresh corn appearance assessment. However, the approach requires Vis-NIR imaging, which increases system complexity and cost. In contrast, Maisagip utilizes

RGB images captured through conventional cameras or mobile devices to provide a more accessible solution for farmers and agricultural stakeholders.

Local Studies

Although artificial intelligence and computer vision applications in Philippine agriculture are still developing, several local studies have demonstrated the potential of deep learning-based approaches for agricultural monitoring and assessment. However, most existing Philippine studies focus on crop disease detection, crop monitoring, and production-related applications rather than post-harvest quality inspection of harvested corn ears.

Malubay and Buban (2024) developed a deep learning-based mobile application for classifying rice crop diseases in Labo, Camarines Norte. The study utilized convolutional neural networks (CNNs) and image processing techniques to classify rice diseases from captured images. The researchers demonstrated that deep learning models combined with smartphone-based image acquisition can provide accessible agricultural decision-support tools for farmers. This study supports the potential of image-based artificial intelligence applications in Philippine agriculture. However, the research focuses on disease identification during crop production rather than post-harvest quality assessment of agricultural commodities.

Navales et al. (2025) developed AgroAI: Leaf-Based Crop Disease Detection System, a web and mobile-based application that uses convolutional neural networks, computer vision, and data analytics for crop disease identification. The system allows users to upload plant images and provides automated disease detection support for crops including corn. The study demonstrated

the practicality of deploying AI-based agricultural applications for local farming communities. However, the system focuses on disease detection from plant leaves and does not evaluate harvested agricultural products, visible defects, or quality grades required for post-harvest inspection.

Recent Philippine initiatives have also explored AI-powered agricultural decision-support systems. For example, the Department of Science and Technology (DOST) and Mountain Province State University developed ClimaPestDetect, an AI-powered platform designed for pest and disease identification and agricultural management support. Such developments demonstrate the increasing adoption of artificial intelligence in Philippine agriculture. However, these systems primarily address crop health monitoring and production challenges rather than automated quality grading of harvested agricultural commodities.

Despite these developments, few Philippine studies have investigated computer vision-based post-harvest quality inspection of white corn and yellow sweet corn ears intended for human consumption. Existing local research mainly focuses on crop disease detection, monitoring, and production improvement, creating a research gap in automated grading, defect detection, and quality assessment after harvesting. Therefore, Maisagip addresses this gap by proposing a hybrid deep learning framework that integrates corn variety classification, visible defect detection, quality grading, physical characteristic estimation, and automated reporting using RGB images.

Synthesis

The reviewed literature and related studies demonstrate that computer vision and deep learning have become important in agricultural quality inspection by providing accurate, efficient, and non-destructive methods for evaluating agricultural products through digital image analysis. Recent literature emphasized that computer vision enables objective assessment of observable physical characteristics, while deep learning architectures automatically learn complex visual features related to color, texture, shape, and surface defects without requiring manual feature extraction. The studies further highlighted that integrating multiple artificial intelligence techniques can improve inspection accuracy, consistency, and practical usability in real-world agricultural applications.

The literature further indicates that hybrid deep learning frameworks offer advantages over conventional single-model approaches because they allow different models to perform specialized tasks within a unified intelligent system. Image classification models are effective for identifying crop categories, object detection models excel in locating visible defects, and rule-based decision support systems provide standardized and explainable decisions based on established grading criteria. Moreover, the adoption of Philippine National Standards (PNS/BAFS) provides an objective basis for quality assessment by defining observable characteristics that influence post-harvest grading and product utilization.

The reviewed foreign studies demonstrated the successful application of deep learning techniques such as convolutional neural networks, YOLO-based object detection models, semantic segmentation networks, and advanced feature extraction architectures for maize defect detection, kernel classification, seed inspection, and fresh corn appearance assessment. However,

most existing systems perform only a single inspection task, focus primarily on maize kernels or seeds, or require specialized imaging technologies such as hyperspectral or near-infrared imaging, which increase implementation cost and reduce accessibility for local agricultural stakeholders. On the other hand, the local studies showed the growing adoption of artificial intelligence and computer vision in Philippine agriculture but were mainly limited to crop disease detection, crop monitoring, and production-related applications, revealing a lack of research on automated post-harvest quality inspection of harvested corn ears intended for human consumption.

Based on these findings, a clear research gap exists in the development of a practical, RGB image-based intelligent system capable of performing comprehensive post-harvest quality inspection of both white corn and yellow sweet corn ears within a single framework. Existing studies rarely integrate corn variety classification, visible defect detection, physical characteristic estimation, quality grading, utilization recommendation, and automated inspection reporting into one system. Furthermore, limited Philippine research has explored the application of computer vision for standardized post-harvest corn quality assessment based on recognized national grading principles.

To address these gaps, the present study proposes “Maisagip: A Hybrid Deep Learning Framework for Corn Ear Quality Grading and Defect Detection Using Computer Vision.” The proposed system integrates EfficientNetV2 for corn variety classification, YOLOv11s for visible defect detection, and a Rule-Based Decision Support System based on PNS/BAFS grading principles to provide objective, consistent, and efficient quality assessment using conventional RGB images. By combining these complementary components within a mobile-responsive web application, the study aims to improve the accuracy, reliability, and practicality of post-harvest

quality inspection while supporting more standardized grading and decision-making for farmers, agricultural cooperatives, traders, retailers, and other stakeholders in the Philippine agricultural sector.

References

Bureau of Agriculture and Fisheries Standards. (2022). *PNS/BAFS 98:2022 – Sweet Corn — Product Standard — Grading and Classification*.

https://mwdjnvdtxvxcheksfsht.supabase.co/storage/v1/object/public/standards-files/2022/1749993020689_PNS_BAFS%2098_2022%20-%20Sweet%20Corn%20-%20Product%20Standard%20-%20Grading%20and%20Classification.pdf

Bureau of Agriculture and Fisheries Standards. (2018). *PNS/BAFS 15:2018 – Corn (Maize) Grits — Grading and Classification*.

https://drive.google.com/file/d/1pebmVT7c-D8-IYhL-m1g6Y_INfBSACnR/view

Cao, Z., Sun, S., & Bao, X. (2025). *A Review of Computer Vision and Deep Learning Applications in Crop Growth Management*. *Applied Sciences*, 15(15), 8438.

<https://www.mdpi.com/2076-3417/15/15/8438>

Chen, R., Yang, M., Qi, H., Luo, X., Liang, Y., Chen, Z., Jian, T., Jiang, J., Huang, H., & Ouyang, J. (2026). *Post-harvest sorting of fresh corn based on semantic segmentation*. *Journal of Agriculture and Food Research*, 26, 102656.

<https://www.sciencedirect.com/science/article/pii/S2666154326000268?via%3Dihub>

Kamilaris, A., & Prenafeta-Boldú, F. X. (2023). *A review of deep learning techniques used in agriculture. Ecological Informatics*, 77, 102217.

<https://www.sciencedirect.com/science/article/abs/pii/S1574954123002467?via%3Dihub>

Kostopoulos, G., Davrazos, G., & Kotsiantis, S. (2024). *Explainable Artificial Intelligence-Based Decision Support Systems: A Recent Review. Electronics*, 13(14), 2842.

<https://www.mdpi.com/2079-9292/13/14/2842>

Liao, B., Wang, Y., Li, X., Cheng, Y., Zhao, G., & Zhou, Y. (2026). *Deep Learning-Based Image Recognition for Food Science and Technology: End-to-End Workflows and Domain-Specific Solutions. Comprehensive Reviews in Food Science and Food Safety*, 25(1), e70388.

<https://ift.onlinelibrary.wiley.com/doi/10.1111/1541-4337.70388>

Malubay, J. C., & Buban, M. G. A. (2024). A deep learning-based mobile application for classifying rice crop diseases in Labo, Camarines Norte. *International Journal on Recent and Innovation Trends in Computing and Communication*, 11(9), 4982–4986.

<https://ijritcc.org/index.php/ijritcc/article/view/10153>

Navales, E. C., Manaog, J., Badajos, X. Y. S., & Cahilig, A. (2025). *AgroAI: Leaf-based crop disease detection for Brgy. Lidong farmers using convolutional neural networks, computer vision, and data analytics. International Journal of Research and Innovation in Applied Science*, 10(11), 57–68.

<https://rsisinternational.org/journals/ijrias/view/agroai-leaf-based-crop-disease-detection-for-brgy-lidong-farmers-using-convolutional-neural-networks-computer-vision-and-data-analytics>

Nawaz, U., Zaheer, M. Z., Khan, F. S., Cholakkal, H., Khan, S., & Anwer, R. M. (2025). *AI in agriculture: A survey of deep learning techniques for crops, fisheries and livestock*. *arXiv*.

<https://arxiv.org/abs/2507.22101>

Pai, D. G., Balachandra, M., & Kamath, R. (2025). *Explainable AI in Agriculture: Review of Applications, Methodologies, and Future Directions*. Engineering Research Express.

<https://iopscience.iop.org/article/10.1088/2631-8695/ae086a>

Tan, M., & Le, Q. V. (2021). *EfficientNetV2: Smaller models and faster training*. In *Proceedings of the 38th International Conference on Machine Learning* (pp. 10096–10106).

<https://arxiv.org/abs/2104.00298>

Wang, B. (2026). *A review of machine vision and deep learning applications in grain quality inspection*. *Emirates Journal of Food and Agriculture*.

<https://ejfa.pensoft.net/article/175268/>

Xia, Y., Che, T., Meng, J., Hu, J., Qiao, G., Liu, W., Kang, J., & Tang, W. (2024). *Detection of surface defects for maize seeds based on YOLOv5*. *Journal of Stored Products Research*, 105, 102242.

<https://www.sciencedirect.com/science/article/abs/pii/S0022474X23001686?via%3Dihub>

Xue, Y., Li, J., & Chen, T. (2026). *Applications of image recognition in intelligent agricultural engineering: A comprehensive review*. *Agriculture*, 16(5), 496.

<https://www.mdpi.com/2077-0472/16/5/496>

Xu, R., Hua, C., Yu, J., & Chen, Y. (2026). *Fresh corn appearance detection using Vis–NIR fusion and FCD-GELAN*. *Journal of Food Process Engineering*, 49(3), e70424.

<https://onlinelibrary.wiley.com/doi/10.1111/jfpe.70424>

Yang, L., Liu, C., Wang, C., & Wang, D. (2024). *Maize kernel quality detection based on improved lightweight YOLOv7*. *Agriculture*, 14(4), 618.

<https://www.mdpi.com/2077-0472/14/4/618>